

PROJECT SCOPE

Meeting Room Booking & Management System

A centralized web-based meeting room booking and management system for organizing rooms, bookings, participants, facilities, equipment, approvals, and administrative operations.

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1. Project Overview

MeetSpace Enterprise Suite is a centralized web application designed to simplify the management of meeting rooms and related workplace resources. The system provides a single interface through which authorized users can view room availability, create and manage bookings, add participants, manage facilities and equipment, and perform administrative operations according to their assigned role.

The project focuses on the core operational workflow: secure access → resource discovery → booking → approval/management → participant and resource management → monitoring and auditability.

The solution is responsive and uses a role-based architecture that separates ordinary user actions from administrative functions.

Core room-booking functionality is the primary deliverable. Microsoft 365/Teams integrations are treated as future or advanced integration work rather than completed functionality.

2. Background and Problem Statement

Organizations may manage meeting rooms through fragmented calendars, spreadsheets, chat messages, or manual coordination. This can result in limited visibility of room availability, duplicate bookings, inefficient resource allocation, and difficulty tracking booking ownership and approvals.

MeetSpace addresses these problems through centralized room data, availability information, conflict prevention, participant management, approval workflows, administrative controls, and audit records.

3. Project Objectives

- Provide a centralized platform for meeting-room discovery and booking.
- Allow users to view room availability and relevant room facilities before booking.
- Prevent conflicting or overlapping bookings for the same room.
- Support participant management and recurring meeting workflows.
- Provide approval and status tracking for controlled bookings.
- Provide role-based access control for administrative and operational functions.
- Manage rooms, locations, departments, facilities, equipment and users through appropriate interfaces.
- Provide secure authentication, password recovery, email verification and protected password handling.
- Improve usability through responsive UI, AJAX interactions, searchable Select2 controls and

server-side pagination.

- Maintain auditability of important administrative and booking-related operations.

4. Scope of the Project

Area	Scope	Status
Authentication	Registration, login, email verification, forgot/reset password and secure password handling.	Implemented
Users & Roles	Users, roles, role assignment and protected administrative operations.	Implemented
Locations & Rooms	Locations, rooms, room details, capacity and facilities.	Implemented
Bookings	Create, view, update and manage room bookings with conflict prevention.	Implemented
Participants	Add and manage participants and participation status.	Implemented
Approvals	Booking approval/status workflow for controlled operations.	Implemented
Equipment & Facilities	Manage facilities and bookable equipment/resources.	Implemented
Audit & Administration	Settings, audit logs and controlled management interfaces.	Implemented
QR Check-in	Room check-in workflow using room QR access.	Implemented
Teams / Outlook	Microsoft 365 calendar and Teams online-meeting integrations.	Future / Advanced

5. Functional Requirements

a) Authentication and Account Management

- User registration with client-side validation and secure server-side processing.
- Email verification using an OTP/verification workflow.
- Login and authenticated session management.
- Forgot-password workflow with email delivery.
- Password reset with immediate frontend validation for required fields, minimum length, complexity and confirmation matching.
- Passwords are securely hashed rather than stored as plain text

b) User Management

- Authorized administrators can create, view, update and manage users.



- Users can be associated with departments and roles.
- Administrative operations are protected by role-based authorization.
- Privilege escalation through unauthorized role assignment is restricted.

c) Role and Access Management

- Role management is available through an administrative interface.
- The system supports role-based access for different user categories.
- Protected operations are checked server-side before sensitive changes are accepted.
- Assigned-user information can be viewed for roles.

d) Department and Location Management

- Manage organizational departments.
- Manage office/location records.
- Associate rooms with locations.
- Restrict administrative mutations according to authorization rules.

e) Room Management

- Create, view, update and manage meeting rooms.
- Store room capacity, location and identifying information.
- Associate facilities with rooms.
- Display room availability information for booking workflows.

f) Facilities and Equipment

- Maintain facility records such as conferencing, presentation and writing resources.
- Manage equipment records and availability/assignment.
- Associate relevant resources with bookings.

g) Booking Management

- Create room bookings with meeting title, date/time, room and organizer info.
- Validate booking data before processing.
- Check room and user references.
- Prevent overlapping/conflicting bookings.
- Support booking updates and cancellation/deletion subject to authorization.



- Display booking status and approval information.
- Support calendar-style booking visualization.

h) Participants and Recurring Meetings

- Add participants to bookings.
- Manage participant records and response/status information.
- Support recurring meeting workflows within the implemented booking.

6. Approval and Operational Workflow

Stage	Description
1. Discovery	User searches/views rooms, availability, capacity, facilities and relevant resources.
2. Booking Request	User enters meeting information and selects the required room and resources.
3. Validation	The application validates the request and checks for conflicting room bookings.
4. Approval / Status	Bookings can follow the configured approval/status workflow.
5. Participant Management	Participants can be added and participation information maintained.
6. Meeting / Check-in	Users can use booking and room check-in functionality during the operational workflow.
7. Update / Cancel	Authorized users can update or cancel bookings according to access rules.

7. Non-Functional Requirements

Requirement	Description
Security	Authentication, authorization, CSRF protection, secure password hashing, controlled administrative access and auditability.
Usability	Clear forms, inline validation, searchable controls, responsive layouts and consistent navigation.
Performance	AJAX retrieval and server-side pagination reduce the amount of data loaded for large listings.
Maintainability	Modular CodeIgniter controllers, models, views and JavaScript modules support continued development.
Responsiveness	Bootstrap-based responsive interfaces support common desktop and mobile browser sizes.



Reliability

Validation and conflict checks help prevent invalid or inconsistent operational data.

8. Technology Stack

Technology	Purpose
PHP 8.x	Application/server-side programming language.
CodeIgniter 4	Backend MVC framework and routing/application structure.
MySQL 8	Relational database management.
Bootstrap 5	Responsive UI components and layout.
JavaScript / AJAX	Client-side interactions and asynchronous data loading.
Select2	Searchable and AJAX-enabled selection controls.
Brevo SMTP	Transactional email delivery for verification/password recovery workflows.
Git / GitHub	Version control and source-code management.

9. System Architecture

MeetSpace follows a web application architecture based on CodeIgniter 4's MVC structure. Browser requests are routed to controllers, which coordinate validation, authorization, models and database operations. Views and JavaScript provide the user interface, while AJAX endpoints supply dynamic data for searchable and paginated screens.

Layer	Responsibilities
Presentation	PHP views, Bootstrap styling, responsive components, forms, dashboards and client-side validation.
Client Interaction	JavaScript/AJAX, Select2, dynamic tables, modals, filters and asynchronous API requests.
Application	CodeIgniter controllers, routing, validation, authorization and workflow logic.
Data	CodeIgniter models and MySQL tables for users, bookings, rooms, locations, roles and supporting modules.
External Services	SMTP email delivery. Microsoft 365/Teams integration is reserved for future/advanced implementation.

10. Database Scope

Module / Table	Purpose
users	User accounts and organizational associations.
roles	Role definitions and access classification.
departments	Organizational departments.



locations	Office/location information.
rooms	Meeting-room information and location relationships.
facilities / room_facilities	Facilities and room-to-facility relationships.
bookings	Meeting-room reservations and booking status.
booking_participants	Participants associated with bookings.
booking_resources / equipment	Resources and workplace equipment associated with bookings.
booking_checkins / booking_visitors	Check-in and visitor information.
settings	Application configuration values.
audit_logs	Recorded audit/activity information for traceability.
migrations	CodeIgniter database migration history.

11. User Roles and Access Control

MeetSpace uses role-based authorization so sensitive operations are not available to every authenticated user.

Permissions are enforced by the application rather than relying only on hiding interface elements.

Role	General Scope
Admin	Full administrative management, including users, roles, settings and protected operational modules.
Manager	Management-level booking and operational access according to configured permissions.
Team Lead	Team-related booking and participant workflows within assigned permissions.
Employee	Standard booking, participant and room-resource workflows.
HR	HR-related organizational access within configured permissions.
Facilities Manager	Facilities, rooms, equipment and relevant booking-management operations.
Intern	Restricted user access based on configured permissions.

12. Security Scope

- Authenticated access is required for protected application areas.
- Sensitive administrative operations use role-based authorization checks.
- Booking modification/deletion is protected against unauthorized access to another user's booking.
- Participant management is protected against unauthorized modification.
- User role assignment is restricted to authorized administrative operations.



- Audit log access is restricted to authorized users.
- CSRF protection is used for applicable state-changing operations.
- Passwords are securely hashed and are not stored as plain text.
- Frontend validation provides immediate feedback while backend security controls remain in place for untrusted requests.
- Database relationships and foreign-key integrity are maintained for implemented modules.

13. Search, Pagination and Data Management

The project includes server-side pagination, search and filtering for major listing modules. The backend uses LIMIT/OFFSET-style retrieval and returns pagination metadata through JSON endpoints, while the frontend requests the required page dynamically.

Searchable Select2 controls are used where relational datasets benefit from asynchronous lookup. This reduces unnecessary data loading and provides a practical user experience for rooms, users, bookings, departments, equipment and related selections.

14. Notifications and Email

Transactional email is configured through Brevo SMTP for workflows such as email verification and password recovery. SMTP credentials are maintained through environment configuration and are not intended to be committed to source control.

15. Testing and Quality Assurance

The project was verified through backend, frontend, browser and regression testing. Testing covered major functional areas and security-sensitive workflows.

Test Area	Coverage / Result
Authentication	Registration, login, email verification, forgot/reset password and validation workflows verified.
Authorization	Role-based access and identified IDOR/privilege-escalation cases addressed and security-tested.
Booking	CRUD operations, validation, overlap/conflict prevention and authorization tested.
Participants	Participant CRUD and access-control scenarios tested.
Pagination	Server-side pagination/search/filter behavior verified across major listing modules.
Select2 / AJAX	AJAX lookup and searchable selection controls verified, including cascading selections.



MeetSpace

Enterprise Suite

UI	Dark/light themes, responsive pages, animations and major user-facing screens reviewed.
Email	Brevo SMTP connectivity and live password-recovery email delivery verified.
Database	Database integrity, relationships and orphan-record checks verified during final testing.

16. Project Deliverables

- Working MeetSpace Enterprise Suite web application.
- Source code managed through Git/GitHub.
- MySQL database schema and application migrations.
- Authentication and account-management workflows.
- Room, location, facility and equipment management.
- Booking, participant and approval workflows.
- Administrative roles, settings and audit functionality.
- Responsive interface with light/dark theme support.
- Server-side pagination, search/filtering and AJAX/Select2 controls.
- Transactional email configuration for verification and password recovery.
- Project documentation, including this Project Scope Document and User Manual.

17. Limitations and Assumptions

- The project is primarily focused on the core meeting-room booking and management workflow.
- Microsoft Teams and Outlook/Microsoft 365 integrations are not represented as completed features in this document.
- External Microsoft integrations may require an eligible Microsoft 365 tenant, permissions and organizational configuration.
- Email delivery depends on valid SMTP configuration and service availability.
- Administrators are expected to maintain accurate users, rooms, locations, facilities and equipment data.

18. Future Enhancements

- Microsoft Teams automatic online-meeting creation and meeting-link synchronization.
- Microsoft Outlook / Microsoft 365 calendar synchronization.
- Microsoft Graph integration and enterprise SSO/MFA.
- Advanced calendar synchronization and external invitation management.
- Enhanced analytics, room utilization dashboards and reporting.



- Expanded visitor and reception workflows.
- Additional notification channels and enterprise integrations

19. Conclusion

MeetSpace Enterprise Suite establishes a centralized foundation for meeting-room booking and workplace resource management. The implemented system covers the primary operational lifecycle from authentication and room discovery through booking, approval, participant management, resource allocation, check-in and administration.

The architecture is modular and designed to support future integrations without making those integrations a prerequisite for the core booking workflow. This provides a practical base for continued enhancement while keeping the current system focused, testable and maintainable.

Item	Details
Core Deliverable	Meeting-room booking and management platform
Primary Stack	PHP • CodeIgniter 4 • MySQL • Bootstrap 5 • JavaScript/AJAX
Project Status	Core system implemented and demonstrated
Document	Project Scope Document • October 2026